Building Circuits – Week 8

Record all measurements made as part of the lab under the relevant section. Graphs of current or voltage versus resistance when relevant are encouraged.

Set up your breadboard the same way you did last week: connect one of the 5V pins on the RPi to the + column and one of the ground pins to the – column. Run connectors from the + and – columns to rows on the main part of the breadboard.

1. LED in a circuit

An LED (light emitting diode) is a component that only allows current to flow in one direction, and it emits light when current passes through it. Like all diodes, it has a positive side (called the anode, the longer wire) and a negative side (called the cathode, the shorter wire).

a. Build a circuit with a resistor of at least 150Ω and an LED in series, connected between the +5V and ground rows.

b. How does the diode need to be oriented? Which wire on the LED goes to the +5V side and which goes to the GND connector?

The longer side needs to go into the positive side

c. What is the voltage drop across the resistor? Was this what you expected?

1.8V. This was not what we expected, as this means that the LED voltage drop is quite significant

d. What is the voltage drop across the LED?

3V

e. Try removing the resistor from the circuit, keeping the circuit closed – the LED is just in series with the 5V supply.

f. What do you think will happen to the LED brightness?

NOTE: Do not leave the LED in this circuit closed for very long.

The LED became very bright

g. What important purpose does the resistor serve?

It limits the current flowing through the circuit so that the wires and equipment does not overheat

h. Try including resistors of different values – how does LED brightness change vs resistor strength?

Among the tested resistors, we found that higher resistance reduced the LED's brightness. This makes sense because a higher resistance limits the current going through the system which limits the power available to the LED, which causes it to be dimmer.

i. Measure the voltage drop across the resistor and LED in each case and make note of voltage drop versus resistance.

Voltage difference across light bulbs drops with increased resistance while voltage difference across resistors increases.

47 kOhms:

- Resistor drop 3 V
- LED 2.18 V

4.7 kOhms

- Resistor drop 2.84V
- LED Drop 2.34V

150 Ohms

- Resistor drops 1.8V
- LED drops 3V

j. Using the configuration with the highest LED brightness, move the 5V connection on the RPi to one of the 3.3V pins. What do you expect to happen to the LED brightness?

We expected the brightness to decrease because current is proportional to voltage and if the voltage drops, then there will be less current which means less power being used by the LED which results in a lower brightness

k. Move the RPi voltage connection back to 5V.

2. Boost converter (step-up voltage)

A boost converter is a small circuit that takes a lower input voltage and converts it to a higher output voltage. You will use one to take the 5V from the RPi and increase it. This is the same kind of component we will use in a future lab to provide the higher voltage needed to power a radiation detector.

l. Add your step-up circuit component to increase your RPi voltage from 5V. Do not close your circuit yet.
m. Use the voltmeter to examine the output voltage of the step-up component and dial it up or down until the output is above 5V but below 10V.
n. Using the dimmest LED configuration from before (meaning select the appropriate resistor from those you tried previously), connect the LED circuit to the boost converter output instead of the RPi 5V directly.
o. How does the LED brightness change as you increase the output voltage?

With a 47kOhms resistor, we saw as we increased the voltage that it was getting brighter

p. How does the voltage drop across the resistor and LED change?

8.3V

- Resistor change is 6.1V, LED is 2.2V

8.0V

- Resistor change is 5.8V, LED is 2.1V

7.1V

- Resistor change 4.9V, LED 2.1V

LED seems to stay the same, while the resistor voltage drop decreases with decreasing overall voltage

q. Do any of these results change with different color LEDs? Specifically, do any voltage drop values change? Is the relative brightness similar for different color LEDs?

For the yellow LED at 7.1V, the brightness was much lower than the green one

- Voltage drop across resistor 5.3V, LED is 1.7V
- The yellow LED has less voltage drop

For the blue LED at 7.1V, the brightness was decent but not as bright as green

- Resistor was 4.6V, LED was 2.4V

NOTE: The boost converter works by rapidly switching current on and off internally. This creates a steady higher voltage on the output, but it also produces small, fast fluctuations called "ripple" or "noise" riding on top of that steady voltage. For powering an LED this does not matter, but in a future lab when we use a boost converter to power a sensitive detector, this noise will become very important. Keep this in the back of your mind.

3. Photo-diode

A photo-diode is a type of diode that generates a small electrical current when light hits it. Unlike the LED (which produces light from current), the photo-diode does the opposite – it converts light into current. Photo-diodes are used in "reverse bias" mode, meaning you connect the voltage backwards compared to how you connected the LED. In reverse bias, very little current flows normally, but when light hits the diode, it creates a small current that is proportional to the amount of light.

a. Replace the LED with a photo-diode. Start with the step-up component outputting 5V.

   NOTE: Photo-diodes operate in reverse bias mode so you will need to orient the diode the opposite way from how you had the LED.

b. What is the voltage across the resistor when you have a 5V bias and the photo-diode is exposed to the ambient light in the room?

With 49kOhms

- Resistor drop is 0.7V

c. What happens if you cover the photo-diode with your hand? What happens if you increase the bias voltage to 10V?

- When we cover the diode, the resistor voltage drop decreases
- When the voltage is set to 10, the resistor drop is still 0.7V

d. What is the "dark current" for this photo-diode? Dark current is the tiny amount of current that flows through the diode even when no light is hitting it. Use the voltage across the resistor and Ohm's law to calculate the current flowing through the diode.

- With the diode fully covered, the resistor measures 0.01V
- Implies the current = V/R = 0.01V/(47000) = $2.128*10^{-7}$ A

e. Is 5V enough supply voltage to see a signal from this diode?

- When a light is shined on the photodiode, we see a large voltage drop across the resistor which indicates that the photodiode is generating current

f. Try shining the flashlight from your phone onto the diode. What is the current output? Again, use the voltage across the resistor to determine this.

- When the light is shined, the drop across the 47kOhms resistor is 4.94V
- This implies current is 0.105mA

g. Are you able to determine the saturation current for the photo-diode? Saturation current is the maximum current the diode will produce no matter how much more light you add.

- We were unable to saturate it, as the drop across the resistor always reached what the system voltage was, meaning that no more current was being generated purely because of the system setup.

4. Looking ahead

The photo-diode you just worked with produces a current when it detects light. Next week, you will work with a similar kind of sensor – a PIN diode – that is sensitive enough to detect individual flashes of light produced by radiation interacting with a crystal. The key challenge you will face is that the signal from a bare diode like this is extremely small. You will use an amplifier circuit to make that signal large enough to see on an oscilloscope. You will also use a boost converter like the one from today's lab to power a radiation detector, and the filtering concepts from last week's simulation will become directly relevant for getting a clean signal.